INARA: Intelligent exoplaNet Atmospheric RetrievAl

A Machine Learning Retrieval Framework with a Data Set of 3 Million Simulated Exoplanet Atmospheric Spectra

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Introduction: Recent advances in machine learning (ML) and computer vision (Krizhevsky et al., 2012; He et al., 2016) offer new ways to reduce the time needed to perform atmospheric retrievals by orders of magnitude (Marquez-Neila et al., 2018; Zingales & Waldmann, 2018), given a sufficient data set to train with. Such advances are becoming increasingly important as the study of exoplanets shifts from their detection to the characterization of their atmospheres, and ultimately to the determination of extant life through biosignatures (Schwieterman et al., 2018). Here we present a ML-based retrieval framework called Intelligent exoplaNet Atmospheric RetrievAl (INARA) that consists of a Bayesian deep learning model for atmospheric retrieval and a data set of 3,000,000 simulated atmospheric spectra of rocky, terrestrial exoplanets generated using the NASA Planetary Spectrum Generator (Villanueva et al., 2018).

Approach: We first generated a spectral data set based on a given planetary system model, where we consider F-, G-, and K-type main sequence stars. Planetary parameters (radius, mass, surface pressure, semimajor axis, pressure temperature profile) were randomly selected from ranges and distributions consistent with our solar system and observations of other systems (Boyajian et al., 2013; Robinson & Catling, 2014; Rogers, 2015; Sotin et al., 2007; Zahnle & Catling, 2017). We consider 12 molecules with concentrations based on the composition of atmospheres in our solar system, as well as the observability of species (Fujii et al., 2018). While cloud mixing ratios were calculated, clouds were ignored at this stage in our simulations due to the computational burden. Observations were simulated using an instrument model of LUVOIR-A, but at a much higher resolution.

INARA is capable of data generation, as well as training, validating, and testing ML model(s) on the fly. Using INARA, we performed a model grid search to determine a suitable ML model – currently, a 1D Convolutional Neural Network (CNN). We then trained the 1D CNN in a supervised setting to predict exoplanet atmospheric parameters given an observed spectrum. We included Monte Carlo dropout in our model to produce predictive distributions over parameters of an exoplanet given a spectrum (Gal & Ghahramani, 2016).

Results and Discussion: Initial results from the best performing 1D CNN model trained using 110,000 parameter–spectra pairs out of the 3M data set demonstrate the capability of INARA to retrieve relatively accurate molecular concentrations given an observed spectrum (Figure 1). Additionally, INARA outperforms previous approaches while computing a larger set of parameter and atmospheric molecules (c.f., Marquez-Neila et al., 2018; Zingales & Waldmann, 2018).

While we obtained promising results on the 110k data set, our search for the best performing model is incomplete. A thorough exploration of different neural network architectures and respective hyperparameters is desirable. Further investigation is also necessary to determine how our predictive distribution using Monte Carlo dropout compares to the posterior distributions of traditional methods.

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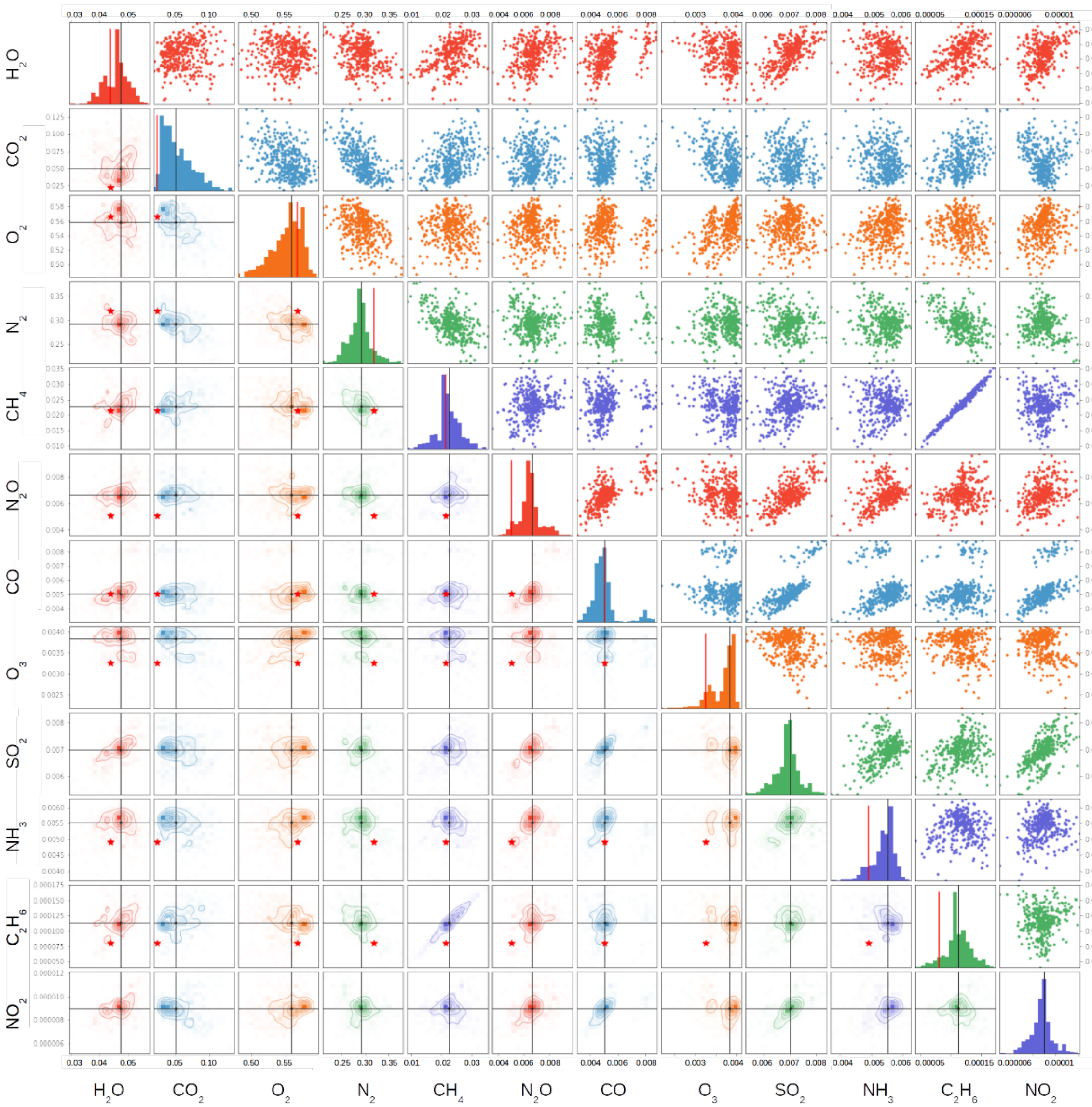

Figure 1. Predictive joint distributions of 12 atmospheric molecules for a random planet among those simulated. The true value, indicated by the red star and the red line, falls within the predictive distribution for many of the molecules retrieved.